Full Stack Assignment – 15

Assignment ID: FSD-15

Title: Complaint Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Complaint Management System** that allows users to raise complaints and support staff to manage their resolution.

The system must enforce resolution flow rules and ensure **data persistence using a database**.

2️⃣ User Roles

USER

- Register and log in to the system
- Raise a complaint
- View own complaint status
- Submit feedback (only after resolution)

SUPPORT

- Log in to the system
- View all complaints
- Update complaint status

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Complaint Handling

- USER can submit complaints with basic details
- Complaints must be stored in the database
- SUPPORT can update complaint status
- Complaint data must be retrieved dynamically

4 Business Rule (Critical)

⚠️ **Feedback can be submitted only after a complaint is marked as CLOSED.**

Any feedback attempt before closure must be rejected.

5 Technical Constraints (Mandatory)

- All complaint and feedback data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6 Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and SUPPORT
- Complaint submission form for USER
- Complaint listing with status updates
- Feedback submission option (only for CLOSED complaints)

UI design is not evaluated; logic and flow are the focus.

7 Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints

- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **state enforcement, role control, and end-to-end integration**.